Google Developer Groups
On Campus · Manipal University Jaipur

THE RAG WORKSHOP

EVENT DETAILS

3 NOVEMBER 2026
RLHC 0104
2:00 PM ONWARDS

What the workshop covers, how it runs,
who is speaking and what you will build.

1 About the workshop

The RAG Workshop is a hands-on session by GDG on Campus, Manipal University Jaipur. It teaches Retrieval-Augmented Generation (RAG): the method that lets a language model answer questions from your own documents instead of guessing from memory.

By the end of the session, every participant leaves with a working chatbot that answers questions about their own PDF files and shows which file and page each answer came from.

2 Event at a glance

ITEM	DETAILS
Event	The RAG Workshop: Master the pipeline. Optimize the retrieval.
Date	Tuesday, 3 November 2026
Time	2:00 PM onwards
Duration	About 100 minutes, followed by questions
Venue	RLHC 0104, Manipal University Jaipur
Organiser	GDG on Campus, Manipal University Jaipur
Format	In person. Slides and live coding in Google Colab
Level	Beginner friendly. Basic Python helps but is not required

3 What you will build

1. Upload one or more PDF files into a Google Colab notebook.
2. Extract the text from every page, using OCR for scanned pages.
3. Split the text into small, overlapping chunks that remember their file and page.
4. Turn every chunk into an embedding: a list of 384 numbers that captures its meaning.
5. Search the chunks by meaning and by keywords to find the best evidence for a question.
6. Send that evidence to a language model, which answers with citations like [S1].

4 What will be taught

PART	TOPICS	SPEAKER
Part 01 **The problem & the pipeline**	Why language models hallucinate. Knowledge cutoff and private data. Fine-tuning compared with RAG. The two RAG pipelines. Reading PDFs, cleaning noisy text, and metadata.	Bhargavi
Part 02 **Chunking & embeddings**	What chunking is and why it is needed. Chunk size and overlap. Tokens compared with characters. What embeddings are and why we need them. Text to vector. Vectors and dimensions. Word, sentence and contextual embeddings.	Augustya
Part 03 **Similarity & retrieval**	Cosine similarity and its formula. How a question is matched against every chunk. Top-k retrieval. The maths of search: multiply, add, sort.	Anwesha
Part 04 **Generation & beyond**	The grounded prompt. Why the order of the context matters. The temperature setting. Building the chatbot. Where baseline RAG breaks. What comes next: vector databases, hybrid search, reranking, evaluation and agentic RAG.	Ishaan

5 How the session is conducted

- **Explain, then build.** Each idea is shown on animated slides, then built step by step in the shared Colab notebook.
- **Live demos.** Speakers use interactive slides for chunking, similarity and retrieval, and run the chatbot live on the projector.
- **A fair comparison.** The same question is asked twice: once to the language model alone, and once with retrieved evidence from the PDFs.
- **Break it on purpose.** Participants change settings to see how a simple RAG system fails, and learn the fix for each failure.
- **Volunteers on the floor.** Volunteers walk the room during every hands-on block to help with errors.

6 Schedule

Times are approximate.

TIME	PART	SPEAKER
2:00 PM	Welcome and setup check	Host
2:03 PM	Part 01 · The problem & the pipeline	Bhargavi
2:25 PM	Part 02 · Chunking & embeddings	Augustya
3:00 PM	Part 03 · Similarity & retrieval	Anwesha
3:20 PM	Part 04 · Generation & beyond	Ishaan
3:45 PM	Questions for all four speakers	Everyone

7 Tools used

TOOL	WHAT IT DOES IN THE NOTEBOOK
Google Colab	Runs the notebook in the browser. Nothing to install.
PyMuPDF	Reads text from each PDF page.
Tesseract OCR	Reads text from scanned pages.
all-MiniLM-L6-v2	Embedding model. Turns text into 384 numbers.
BM25	Keyword search, combined with meaning search.
Groq API	Runs the language model, gpt-oss-20b, with gpt-oss-120b as backup.

8 Who should attend

Any student curious about how AI assistants answer questions from documents. No machine learning background is needed. If you can run a cell in a notebook, you can follow along.

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Rules and what to bring are in the Guidelines document.